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Creep leads ~~to~~ to the dimensional change in Zirc over time.

This can cause creep down towards the fuel, and creep out towards the coolant. Creep down will eventually lead to contact with the fuel, and creep out will increase the gap size ~~and~~ negatively impacting heat transfer. This will ~~also~~ decrease the ductility and increase the strength as well.

- Wanted the stages

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Susceptible material, Corrosive environment, enough time, enough stress

Susceptible material - The zirconium is susceptible to attack by volatile FP gases such as iodine. It combines with iodine in a cycle and moves toward the fuel where it recombines with oxygen, releasing the Iodine to attack the crack face again.

Corrosive environment - Both the oxygen in the coolant and gap, as well as the volatiles in the gap create an environment where corrosion of Zirc can occur.
this is only fuel-site

time - The zirconium's susceptibility to ~~stress~~ SCC increases with time. Initially the crack forms, then propagates, then the crack goes all the way through. More time in the corrosive environment under stress will lead to a through wall crack

- mostly...

Stress - Stresses from the fuel interacting with the cladding lead to SCC. As the pellet expands and cracks, it will have preferential contact points with the clad. These high stress zones lead to crack tip forming.